



Research papers

A physical process and machine learning combined hydrological model for daily streamflow simulations of large watersheds with limited observation data

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ARTICLE INFO

This manuscript was handled by Emmanouil Anagnostou, Editor-in-Chief, with the assistance of Qiang Zhang, Associate Editor

Keywords:

Hydrological modeling
Physically distributed model
Machine learning
Computer vision
Categorization approach

ABSTRACT

Physically distributed hydrological models are effective in hydrological simulations of large river basins, but the complex characteristics of hydrological features limit their application. An easy-to-use and high-efficiency hydrological model is needed for efficient water resource management in practice. Machine learning (ML) based models have the potential to provide fast mapping pathways between meteorological predictors and hydrological responses without detailed descriptions of the corresponding physical processes. However, the extensive data requirements, ignoring of spatial variability and poor performance for extreme flows limit the application of ML models. This study attempts to develop an ML-based hydrological model by combining physically based distributed hydrological model with an artificial neural networks (ANN), computer vision (CV) and a categorization approach (CA). To solve the insufficient training problem, we use a physically distributed hydrological model (GBHM) together with a stochastic rainfall generator to generate a large amount of synthetic data (GBHM-ANN). To improve the extreme flow simulation, we add the categorization approach into GBHM-ANN (GBHM-ANN-CA). To capture the spatial variability of the predictors, we also use a local binary pattern-based computer vision method to form GBHM-ANN-CA-CV model. The effectiveness of the three modeling approaches are demonstrated by synthetic case studies. We finally evaluate GBHM-ANN-CA-CV using the real data from the upper Chao Phraya Basin in Thailand. The results show that the prediction accuracy of our new data-driven model is greatly improved in data-limited watersheds. Specifically, the CV extracted spatial information can improve the robustness of the data-driven hydrological model, and the CA can greatly improve high flow simulations. The combined model yields a satisfactory accuracy for long-term daily streamflow simulations. This study demonstrates the potential of ML-based hydrological models in water resource management, especially in changing environments.

1. Introduction

Physically based hydrological models have been widely used for streamflow simulations and forecasting (Arnold and Fohrer, 2005; Chen et al., 2011; Jia et al., 2001; Refsgaard, 1997; Yang et al., 2002). Among the various kinds of physically based models, fully distributed hydrological models, which can consider the spatial variability of the watershed landscape and its atmospheric forcing, are considered to be the gold standard for hydrological modeling (Chen et al., 2016) and have been applied in ungauged basins (Göttinger and Bárdossy, 2007; Hunukumbura et al., 2012; Wambura et al., 2018). However, with

complex model structures and extensive calculation requirements, physically based distributed hydrological models often have high computational costs and require high levels of hydrological expertise for modelers and users, thus limiting their application in water resource management (Chen et al., 2016; Kratzert et al., 2018; Srivastav et al., 2007; Yaseen et al., 2019; Zhang et al., 2018a; Zhang et al., 2018b).

Machine learning (ML)-based methods can recognize patterns hidden in historical data, and they may provide quick and direct mapping pathways between predictors and hydrological responses without explicit descriptions of the underlying physical processes (Adnan et al., 2019; Kasiviswanathan et al., 2016; Sahoo et al., 2017).

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<https://doi.org/10.1016/j.jhydrol.2020.125206>

Received 31 March 2020; Received in revised form 1 June 2020; Accepted 17 June 2020

Available online 23 June 2020

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Many studies have demonstrated that ML models can outperform other state-of-the-art techniques in natural science fields, including hydrology (AlQuraishi, 2019; He et al., 2019; Kratzert et al., 2018; Kratzert et al., 2019a; Kratzert et al., 2019b; Zhang et al., 2020). Among the various ML models, neural networks are the most widely reported methods for streamflow simulations and forecasting. For example, Hsu et al. (1995) compared the runoff forecasting performance of a physically based model called the Sacramento Soil Moisture Accounting Model (SAC-SMA) to that of a three-layer artificial neural network (ANN) under various flow regimes, and the results showed that the ANN model provided a better representation of the rainfall-runoff relationship. Demirel et al. (2009) compared an ANN model with a process-based semi-distributed model called the Soil and Water Assessment Tool (SWAT) for streamflow forecasting one day in advance, and the comparisons showed that the ANN was more successful than SWAT in peak flow forecasting. Humphrey et al. (2016) combined a Bayesian artificial neural network with a physically based conceptual model called GR4J for monthly streamflow forecasting, and the results show that both the hybrid model and pure ANN model outperformed the GR4J conceptual hydrological model. Recently, deep learning (DL) neural network models, such as long short-term memory (LSTM) networks, have been reported to be suitable for rainfall-runoff process modeling. For example, Kratzert et al. (2019b) improved the standard LSTM architecture and proposed an entity-aware-LSTM (EA-LSTM) for hydrological modeling that could learn catchment similarities and outperformed five physically based hydrological models.

However, current ML-based hydrological models suffer from several drawbacks. First, ML models often require a large amount of training data to obtain robust performance (Kalra and Ahmad, 2009; Kratzert et al., 2019b). This characteristic severely limits the applicability of ML in hydrological simulation and prediction because the majority of streams in the world lack long-term hydrological observations (Goswami et al., 2007; Kratzert et al., 2019a; Sivapalan, 2003). Jiang et al. (2018) developed a computer vision-based data-driven model for daily runoff simulations in a gauged basin, and tested the transfer learning performance in an ungauged basin; their results showed that the overall performance of the model was satisfactory in the ungauged basin, but it could not accurately predict high flows. Kratzert et al. (2019a) added catchment characteristics as the predictor variables of an LSTM model that was trained with data from a large number of gauged basins and explored the capability of this regionally trained model for predictions in ungauged basins. Although the ungauged LSTM model displayed better performance than traditional models, it still required large amounts of data from gauged basins with long-term observations for training. Karpatne et al. (2017) introduced a physics-guided or theory-guided data science and showed that hydrological mechanism-guided methods are effective but need further study.

Moreover, traditional ML models often ignore the spatial variability in the watershed landscape and the related atmospheric forcing, and they aggregate data over the watershed as model inputs (Jiang et al., 2018). However, hydrological responses are often significantly impacted by the spatial patterns of driving forces and terrain characteristics (Chen et al., 2016; Jiang et al., 2018; Wang et al., 2015). Computer vision (CV) algorithms have made it possible to overcome this drawback. CV technology is used to extract information from images, which allows machines to understand images by processing digital signals (Wang et al., 2013). Instead of using the values associated with each pixel, CV uses features to quantitatively describe images in low dimensions. CV has been applied to some areas of Earth science and is reported to be able to effectively extract spatial information from images. For example, to classify crops in very high-resolution remote sensing images, Sun et al. (2020) proposed a method guided by hierarchical perception, a CV-based concept, and the model performed well and displayed high precision. With the help of CV, Ling et al. (2019) developed a convolutional neural network-based super-resolution mapping model that can effectively estimate the subpixel-scale details

of rivers and the wetted width. By combining a convolution neural network and LSTM network, Miao et al. (2019) proposed a statistical downscaling method to improve the precipitation prediction of a general circulation model (GCM) in a monsoon region. Among various CV algorithms, local binary pattern (LBP) is one of the most popular methods used in the field of pattern recognition (Gupta et al., 2020; He and Sang, 2013; Heikkilä et al., 2009; Jiang et al., 2018; Khan et al., 2020). LBP was first developed by Ojala et al. (1996) in a comparative study of texture measures, which is an image operator based on local pixel information and is complementary to the contrast information (Gupta et al., 2020). The LBP can potentially be used to capture the watershed spatial pattern in hydrological modeling.

Third, as statistical methods are purely based on observation data, ML-based hydrological models often provide poor simulations of high flows, which are important in practical applications (Sudheer et al., 2003; Wu et al., 2009; Yang et al., 2019). The frequency of high flows in streamflow time series is relatively low, which may lead to insufficient model training (Yang et al., 2019). Sudheer et al. (2003) proposed a data transformation method based on a statistical model to improve the peak flow estimation with an ANN. Since the underlying mechanisms of runoff generation may be quite different under various flow regimes, a single global ML model often fails to provide satisfactory predictions for both extreme high and low values and even normal values (Hsu et al., 1995; Solomatine and Ostfeld, 2008; Wu et al., 2009). Several researchers have attempted to improve the modeling performance by using categorization models, which means that sub-processes were identified first, and separate models were built for each sub-process. To improve the prediction of high magnitude flows, Sivapragasam and Liong (2005) proposed a method to classify flow into low, medium and high flow and built the support vector machine model for each type of flow. Wu et al. (2009) proposed a crisp distributed support vectors regression (CDSVR) model for monthly streamflow forecasts. They used the Fuzzy C-means clustering technique to split the flow data into three subsets, and then fitted three single support vector regressions to three subsets. Zemzami and Benaabidate (2016) and Tongal and Booi (2018) both tried to improve the ML model performance for peak flow simulations by separating the streamflow into a baseflow and quick response flow. Chu et al. (2019) divided the streamflow time series to different flow regimes by a Fuzzy C-means method and developed data-driven models for each regime to map the nonlinear relationship between the selected predictors and streamflow. The categorization approach (CA) needs to be further studied in peak flow simulations with insufficient observation data.

As a physically based distributed hydrological model, the geomorphology-based hydrological model (GBHM) developed by Yang et al. (1998) has been successfully applied in many regions, such as the Yangtze River (Li et al., 2015), Mekong River (Wang et al., 2016) and Jiulong River (Lu et al., 2018). In this study, GBHM is used to provide sufficient training samples for ML models. With gradient-based optimization schemes, the traditional ANN model may yield premature convergence and become trapped at local optima. In addition, the traditional model is very sensitive to the initial conditions, including the initial weights and biases (Yang et al., 2017; Zanchettin and Luderim, 2007). To overcome these drawbacks, the GA is used in this study to optimize the initial conditions of ANN, which can achieve an optimal solution (Bahrami et al., 2016). Thus, the GA-ANN model is used for the hydrological simulation in this study.

In this study, we develop a physical process and ML combined hydrological model for continuous daily hydrological simulations in data-limited watersheds located in northern Thailand. With rainfall data generated by a stochastic rainfall generator, a physically based distributed hydrological model is used to generate sufficient streamflow samples for training and validation in the data-insufficient basins. The spatial features are extracted from the predictor images by using CV algorithms and are mapped to the hydrological responses by categorization models. We strive to (1) verify the effectiveness of the

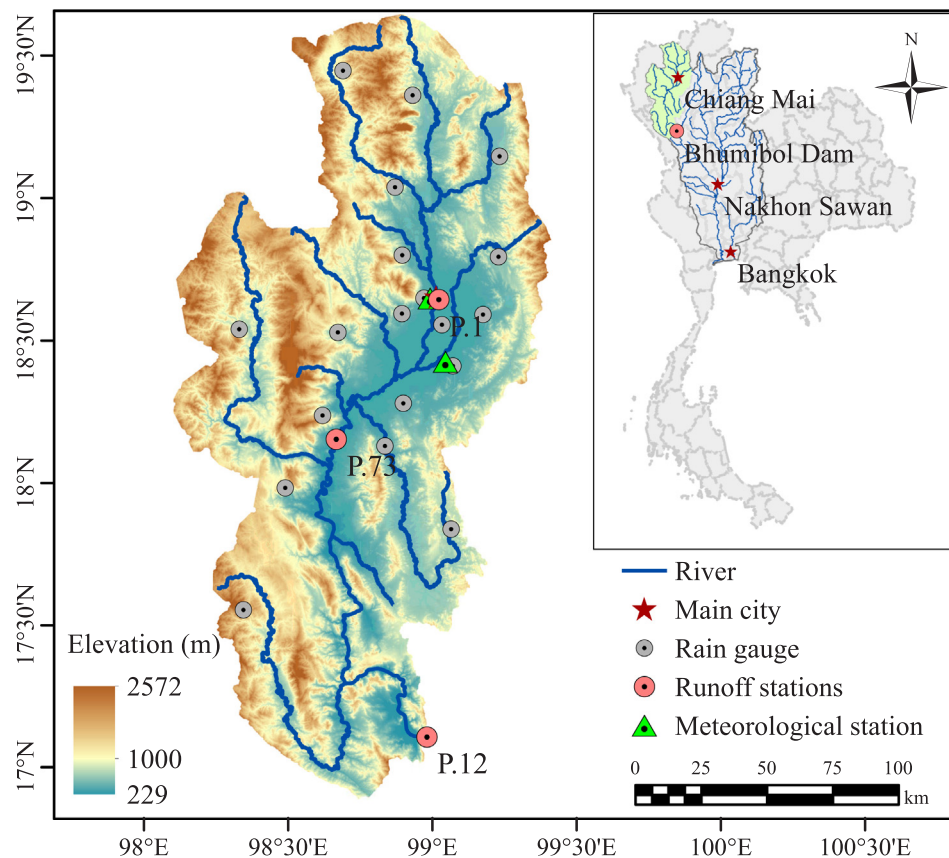


Fig. 1. Study area in the Ping River Basin with three hydrological stations at Nawarat Bridge (P1 station), Chom Thong (P73 station), and Sam Ngao District (P12 station).

distributed hydrological model in supporting the ML-based data-driven model in data-insufficient basins; (2) explore the use of spatial information and predictors extracted by CV; (3) investigate the effectiveness of the categorization approach in improving high flow simulations; and (4) test the applicability of the combined model using real data.

2. Study area and data

2.1. Study area

The study area (see Fig. 1), the Ping River Basin, is located in northern Thailand with a drainage area of 26,386 km²; this basin is upstream of the Bhumibol Reservoir, which is the largest reservoir in Thailand. The Ping River is one of the main tributaries of the Chao Phraya River and flows through Chiang Mai, which is the largest city in northern Thailand. The elevation of the basin ranges from 229 m in the south to 2572 m in the north, and the spatial heterogeneity of the topography and meteorology is significant. The climate is a tropical monsoon pattern with a mean annual precipitation of approximately 900 mm. The temporal distribution of rainfall is highly heterogeneous. On average, approximately 90% of the total annual precipitation is concentrated in the wet season from May to November, and the rest occurs during the dry season from December to April (Ponpang-Nga and Techamahasaranont, 2016). The mean annual air temperature is approximately 22 °C, and forest is the dominant land cover type, which accounts for approximately 80% of the total area.

2.2. Observation data

The data used in this study include daily climate forcing

observations, geographic information from the study area and daily streamflow measurements collected by three runoff gauges. The climate forcing data include daily precipitation, daily air temperature (mean, maximum and minimum), daily mean relative humidity, daily mean wind speed and daily sunshine duration data (Qin et al., 2017; Yang et al., 2019). All climate forcing data, except the wind speed, were observed at 96 rainfall stations and 14 meteorological stations and were provided by the Royal Irrigation Department of Thailand within the Chao Phraya River Basin. These data were spatially interpolated into gridded images with a spatial resolution of 2.5 km × 2.5 km by an angular distance weighting method with elevation corrections (Yang et al., 2004). The spatial distributions of the daily wind speed (0.25° × 0.25°) were obtained from the Global Land Data Assimilation System (GLDAS) (Rodell et al., 2004). To match the spatial resolution of 2.5 km × 2.5 km, we resampled the original data by the nearest neighbor assignment using ESRI ArcGIS 10.6 software. Because the stochastic rainfall generator (SRG) requires more than 20 years of historical records as raw materials, we selected 57 rainfall stations within and around the study area among 96 rainfall stations, which had more than 20 years of data available before January 1, 2010 to generate synthetic data by SRG. The soil data of the Ping River Basin were obtained from the Harmonized World Soil Database (HWSD) (FAO et al., 2012), and the soil water parameters (including the soil hydraulic conductivity, saturated soil moisture and residual soil moisture) were spatially interpolated using the methods proposed by Rawls and Brakensiek (1985). Geomorphological information, including the slope, slope length and elevation, was obtained from the Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) with a spatial resolution of 90 m (Jarvis et al., 2008). The land use maps were provided by the Department of Land Development of Thailand, and the vegetation parameters, including the fraction of photosynthetically

Table 1
Detailed information on the data used in this study.

Data type	Parameter	Original resolution	Unit	Source
Climatic forcing data	Air temperature (mean, maxima and minima)	Daily at 96 rainfall stations and 14 meteorological stations	°C	Royal Irrigation Department of Thailand
	Precipitation		mm	
	Average relative humidity		%	
	Sunshine duration		hour	
	Average wind speed		m s^{-1}	
Geographic information		3-hour and $0.25^\circ \times 0.25^\circ$		GLDAS (Rodell et al., 2004)
	Soil layer depth	One time and $0.00833^\circ \times 0.00833^\circ$	m	HWSD (FAO et al., 2012)
	Soil clay content		% wt.	
	Soil sand content		% wt.	
	Soil organic carbon content		% wt.	
	Land use map	One time and $30 \text{ m} \times 30 \text{ m}$	m	Department of Land Development of Thailand
	Digital elevation model (DEM)	One-time and $90 \text{ m} \times 90 \text{ m}$	m	SRTM (Jarvis et al., 2008)
	Leaf area index (LAI)	15-day and $0.00833^\circ \times 0.00833^\circ$	–	GIMMS (Zhu et al., 2013)
Daily streamflow measurements	Fraction of photosynthetically active radiation (FPAR)			
	Streamflow at gauge P1	Daily	$\text{m}^3 \text{ s}^{-1}$	Royal Irrigation Department of Thailand
	Streamflow at gauge P73			
	Streamflow at gauge P12			

active radiation (FPAR) and the leaf area index (LAI), are obtained from the 3rd-generation Normalized Difference Vegetation Index (NDVI3g) of the Global Inventory Modeling and Mapping Studies (GIMMS) project (Zhu et al., 2013).

Streamflow data at three stations, including the gauge at the Nawarat Bridge (referred to as gauge P1), the gauge in the Chom Thong (referred to as gauge P73) and the gauge in the Sam Ngao District (referred to as gauge P12), were selected to evaluate model performance. The data were provided by the Royal Irrigation Department of Thailand. Gauge P1 is located near Chiang Mai in the upper Ping River Basin, with a drainage area of 6355 km^2 . Gauge P73 is located in the central portion of the Ping River Basin, with a drainage area of $14,814 \text{ km}^2$. Gauge P12 is located at the basin outlet. Detailed information on the data used in this study is given in Table 1.

3. Methodology

3.1. Modeling approaches

To overcome the limitations of previous ML-based hydrological models, we propose three corresponding modeling approaches and then develop a physical process and machine learning combined hydrological model. In addition, we evaluate the effectiveness of the proposed modelling framework by comparing different data-driven models step-by-step.

3.1.1. Three modeling approaches

The main limitations of previous data-driven models in hydrological simulations are the extensive data requirements, ignoring catchment spatial variability and poor performance for extreme flow simulation. Accordingly, we propose three modeling approaches to solve those drawbacks.

To overcome the need for large training data sets, we propose a hydrologically guided modeling approach that can be regarded as a kind of physics-guided data science method (Karpatne et al., 2017). This approach combines a widely used physically distributed hydrological model (a geomorphology-based hydrological model, GBHM) (Yang et al., 1998) with a stochastic rainfall generator (SRG) (Basinger et al., 2010) to generate sufficient physical synthetic data to train ML models. Specifically, the SRG is used to generate synthetic rainfall data, and the calibrated GBHM model is used to generate corresponding runoff data with the synthetic rainfall data set used as the input. This method can avoid the drawback of previous data-driven models and

allow the proposed model to be applied in basins for which long-term hydrological observations are scarce. The details of this method are provided in Section 3.2.

To consider the spatial heterogeneity of predictors, we adopt a local binary pattern-based computer vision method (LBP-CV) (Ojala et al., 1996) to extract the spatial features of predictors and use these spatial features as model inputs. The LBP-CV method is an effective feature representation algorithm, especially for hydrological variables (Jiang et al., 2018), and it can extract both texture and intensity features from gridded images. The details of this method are provided in Section 3.3.

To effectively simulate high flows, we propose a categorization approach (CA), which is similar to the approach introduced by Sivapragasam and Liong (2005). In particular, this approach splits the training data into four sub-datasets (subsets of small, medium, large and extraordinarily large flows) based on the relevant percentiles and then establishes separate ANN models for each sub-dataset. For prediction during the test period, we first use the trained random subspace ensemble k-nearest neighbor (RS-KNN) classifier to predict the flow magnitude labels of streamflow data and then choose the corresponding ANN model to predict the streamflow. The CA can avoid the problem of insufficient training for high flows due to their low frequency in streamflow time series and can thus improve the high flow prediction capability of data-driven models. The details of the ANN and RS-KNN are given in Section 3.4.

3.1.2. GBHM-ANN-CA-CV model

We develop a physical process and ML combined hydrological model (GBHM-ANN-CA-CV) in this study for hydrological simulation, and the modelling framework is shown in Fig. 2. The model mainly consists of four main steps. The first step is data preprocessing, which includes input variable selection, training data generation, and spatial feature extraction. Since this model is driven by GBHM-generated synthetic data, the same inputs are used. The model inputs are composed of three types of variables, including meteorological forcing variables, vegetation parameters and catchment geomorphologic information (Qin et al., 2017; Yang et al., 2002). The meteorological inputs include daily precipitation, daily mean air temperature, daily relative humidity and daily potential evapotranspiration calculated by the Penman-Monteith equation (Monteith, 1965). The vegetation input includes the LAI, which is linearly interpolated to a daily resolution. The catchment geomorphologic inputs include the slope, slope length and elevation derived from the DEM. Since there is a delay from the start of a rainfall event to the time when runoff occurs and this lag time

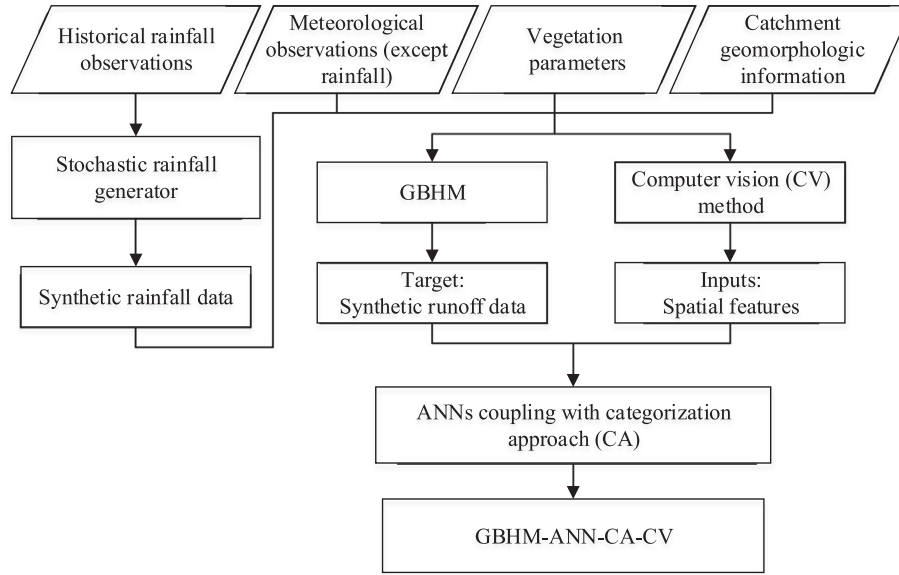


Fig. 2. The framework of the physical process and machine learning combined hydrological model (GBHM-ANN-CA-CV) for daily streamflow simulations, where GBHM represents the geomorphology-based hydrological model, and ANN is the artificial neural network.

Table 2

A summary of input to the physical processes and machine learning combined hydrological model (GBHM-ANN-CA-CV).

Data	Temporal resolution	Spatial resolution	Data source
Precipitation	Daily	2.5 km × 2.5 km	Spatially interpolated from meteorological stations
Mean air temperature	Daily	2.5 km × 2.5 km	Spatially interpolated from meteorological stations
Relative humidity	Daily	2.5 km × 2.5 km	Spatially interpolated from meteorological stations
Potential evapotranspiration	Daily	2.5 km × 2.5 km	Estimated by Penman-Monteith equation
LAI	15-day	0.00833° × 0.00833°	GIMMS
Elevation	One time	90 m × 90 m	SRTM
Slope	One time	2.5 km × 2.5 km	Calculated from DEM
Slope length	One time	2.5 km × 2.5 km	Calculated from DEM

needs to be considered in model development, we calculate the lag time (71.8 h $\approx$ 3 days) of the Ping Basin according to the method suggested by the United States Department of Agriculture (Mockus, 1964), and meteorological predictors with a 3-day lag time are adopted. Detailed information on the model inputs is given in Table 2. After determining the model inputs, the physically based synthetic data for model development are generated by using the SRG and GBHM. With the synthetic dataset, the spatial features of input variables are extracted by LBP-CV and concatenated to form a feature space for ML. To avoid model overfitting, we adopt principal component analysis (PCA) to further reduce the dimensionality of the feature space. PCA is a widely used linear method for data reduction. It tries to identify a small number of uncorrelated variables (i.e., principal components) that are the linear combination of origin variables. The detail information of PCA can be found in Kramer (1998).

We select streamflow at time step t as the response variable for the ML model, and the result can be expressed as

$$Q_t = f(P_t, P_{t-1}, P_{t-2}, P_{t-3}, T_t, T_{t-1}, T_{t-2}, T_{t-3}, H_t, H_{t-1}, H_{t-2}, H_{t-3}, E_t, E_{t-1}, E_{t-2}, E_{t-3}, LAI_t, LAI_{t-1}, LAI_{t-2}, LAI_{t-3}, S, SL, ELE) \quad (1)$$

where Q is streamflow, P is precipitation, T is the mean air temperature, H is the relative humidity, and E is the potential evapotranspiration, which are all functions of time and space. Variable S is the slope, SL is the slope length, and ELE is the elevation; these variables are only functions of space.

The second step is to train and validate the machine learning-based models with the CA. The third step is to predict the streamflow labels

with a random subspace ensemble RS-KNN classifier and generate the predicted streamflow in the test period. The synthetic precipitation and streamflow data are used to develop the RS-KNN classifier. Since precipitation has the most important impact on runoff generation, the areal mean precipitation for the current day and previous three days and the 3-day cumulative areal mean precipitation are selected as the predictor variables for the classifier. The magnitude of the streamflow labels is selected as the target variable. Since the number of measurements with small magnitude labels is far greater than the number of samples with other labels, we simply reproduce the samples with these labels so that the number of samples is approximately equal for all label classes. With the trained RS-KNN, the magnitude labels of streamflow can be predicted, and the fourth step is to choose the corresponding sub-model to predict the streamflow on each day in the test period. This label prediction approach can be expressed as

$$L_t = G(\bar{P}_t, \bar{P}_{t-1}, \bar{P}_{t-2}, \bar{P}_{t-3}, \bar{C}P_t, \bar{C}P_{t-1}, \bar{C}P_{t-2}, \bar{C}P_{t-3}) \quad (2)$$

where L is the label, $\bar{P}$ is the areal mean precipitation, $\bar{C}P$ is the 3-day cumulative areal mean precipitation and G represents the classifier.

We carry out a comparison of three data-driven models for runoff simulations to demonstrate the effectiveness of the modelling framework. The first is a simple ANN with the same predictors as the GBHM-ANN-CA-CV, and it is used as the baseline model. The second is an ANN model driven by GBHM-generated synthetic data (referred to as GBHM-ANN), and the comparison between the ANN and GBHM-ANN can be used to determine if the performance of the ML model driven by GBHM-generated synthetic data is superior for data-insufficient basins. The third is an ANN model coupled with a categorization approach driven by GBHM-generated synthetic data (referred to as GBHM-ANN-CA), and

the comparison between GBHM-ANN-CA and GBHM-ANN can be used to assess the effectiveness of the categorization approach in improving the capability of data-driven models to simulate high flows. The three models all use the areal mean values of predictor variables instead of spatial features. The comparison between GBHM-ANN-CA and GBHM-ANN-CA-CV can be used to explore the effects of spatial variability on the robustness of data-driven hydrological models.

For models driven by GBHM-generated synthetic data, we use 48-year synthetic data for training and 12-year synthetic data for validation. For models driven by observations, we use the observations from 1 January 1990 to 31 December 2005 for training and the observations from 1 January 2006 to 31 December 2009 for validation. The streamflow values at three runoff gauges (i.e., P1, P73 and P12) from 1 January 2010 to 31 December 2016 are predicted by all four models in this study for testing.

We use three different criteria to evaluate the overall performance of the models: the root mean square errors (RMSE), the Nash-Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and the percent bias (PBIAS) (Gupta et al., 1999). NSE is a normalized statistical index with a value between 0 and 1, and a larger value indicates a better simulation. PBIAS reflects the overall flow bias error, and a value close to 0 indicates a good simulation. It is commonly acknowledged that the performance of hydrological models is considered satisfactory if $NSE \geq 0.50$ and $-15\% \leq PBIAS \leq 15\%$ for daily streamflow simulation (Moriasi et al., 2007). The model performance for low flow simulations is typically evaluated based on the logarithmic-transformed Nash-Sutcliffe efficiency (LNSE) (Smakhtin et al., 1998), and the performance of peak flow simulation is evaluated by the peak error (PE), which is just the percent bias of the peak. The peak flow events in streamflow series are identified by the peaks-over-threshold (POT) method, and a threshold that leads to three events per year is adopted in this study. The details of the POT method can be found in the previous study of Mediero et al. (2014).

3.2. Synthetic data by SRG and GBHM

We generate a training and validation dataset by combining the SRG and GBHM to alleviate the limitations of previous ML-based hydrological models in data-insufficient areas. Specifically, we use the SRG to obtain synthetic rainfall data and the GBHM to obtain the corresponding streamflow samples.

3.2.1. Stochastic rainfall generator

We use a nonparametric approach SRG developed by Basinger et al. (2010), which uses a bootstrapped Markov chain to generate synthetic series based on a user-specified historical record of daily rainfall and maintains the statistical characteristics of the historical rainfall. The method includes three major steps. The first step is to calculate the occurrence probabilities of precipitation for each day in historical sequences. The precipitation on each day in the historical series is assigned a “dry” or “wet” label, with “dry” representing a zero value of precipitation and “wet” representing a positive value. To describe the occurrence statistics of historical precipitation, we calculate three types of probabilities for each day in the historical series with the following equations.

$$Prob(W) = \frac{W}{W + D} \quad (3)$$

$$Prob(W|W) = \frac{WW}{WW + WD} \quad (4)$$

$$Prob(D|D) = \frac{DD}{DW + DD} \quad (5)$$

where $Prob(W)$ is the probability of a wet day, $Prob(W|W)$ is the conditional probability of a wet day following a wet day, and $Prob(D|D)$ is the conditional probability of a dry day following a dry day. Letter W

is the number of wet days, D is the number of dry days, WW is the number of wet days following wet days, WD is the number of dry days following wet days, DD is the number of dry days following dry days and DW is the number of wet days following dry days for each day of the year. The second step is to assign “dry/wet” labels to each day in all years of the simulated series considering the seasonality of rainfall. The label for each day is assigned according to the label of the previous day. For a day with a dry label, the simulated precipitation is zero. For a day with a wet label, the amount of rain is randomly selected from the historical precipitation amounts with specific consideration of seasonality. This rainfall generator can ensure the seasonality of synthetic series and avoid the occurrence of unreasonable prediction values. The detailed calculation process was presented by Basinger et al. (2010).

We use the SRG method to generate synthetic rainfall time series for the 57 selected rainfall stations based on their 20-year historical observations. Since precipitation is the main factor in runoff generation, we only generate synthetic rainfall series, and other meteorological variables are still based on historical observations. We obtain corresponding synthetic climate forcing data and spatially interpolate these data to gridded images with a spatial resolution of $2.5 \text{ km} \times 2.5 \text{ km}$ using an angular distance weighting method with elevation corrections (Yang et al., 2004).

3.2.2. Geomorphology-based hydrological model

The GBHM is a distributed hydrological model developed by Yang et al. (1998), and it mainly consists of hydrological simulations in each hillslope unit and river routing along river networks. The simulations in each unit mainly involve calculations of evapotranspiration, canopy interception, surface flow, infiltration, unsaturated flow and groundwater flow. River routing is calculated by kinematic wave methods, and the total runoff at the outlet of each sub-catchment can be obtained by summing the runoff contributions in each grid. Detailed descriptions of the model were given by Yang et al. (1998) and Yang et al. (2002).

We use the GBHM in this study to simulate the hydrological processes in the Ping River Basin. To reflect the real physical situation in the study area, we first calibrate and validate the GBHM using historical observations. The spatial resolution of the model is chosen as $2.5 \text{ km} \times 2.5 \text{ km}$, and the entire basin is split into 58 sub-catchments. The model is activated and initialized with a 6-year warm-up period, and the daily streamflow data at P12 from 1 January 2000 to 31 December 2004 are used to calibrate parameters, including the slope shape factor, groundwater hydraulic conductivity and saturated hydraulic conductivity of the topsoil (Miao et al., 2016). We use the data from 1 January 2005 to 31 December 2009 to validate the model. After calibration and validation, the developed GBHM can be assumed to reflect the real hydrological processes in the region and can be used to generate streamflow samples. Second, with synthetic climate forcing data points as inputs, the calibrated model is used to generate synthetic streamflow series for model development in data-insufficient basins.

3.3. Spatial features extracted by computer vision (CV) algorithms

This study uses CV algorithms to obtain the spatial features of the predictor images. Specifically, the texture feature is extracted by the local binary pattern (LBP) algorithm developed by Jiang et al. (2018), and the intensity feature is represented by the areal mean of pixel values.

For a 3×3 -pixel block, the values of 8 neighboring pixels are first transformed to 0 or 1 using the value at the center pixel as the threshold; then, an 8-bit binary code is indexed for the center pixel. The corresponding decimal of this 8-bit binary code is referred to as the LBP value at the center pixel, and the LBP value of each pixel over an image can then be calculated. To describe the statistical characteristics of the LBP values in an image, a histogram is usually used, and the number of bins in the histograms can be set to range from 1 to 256. In this study, considering the computational complexity, the number of bins is set to

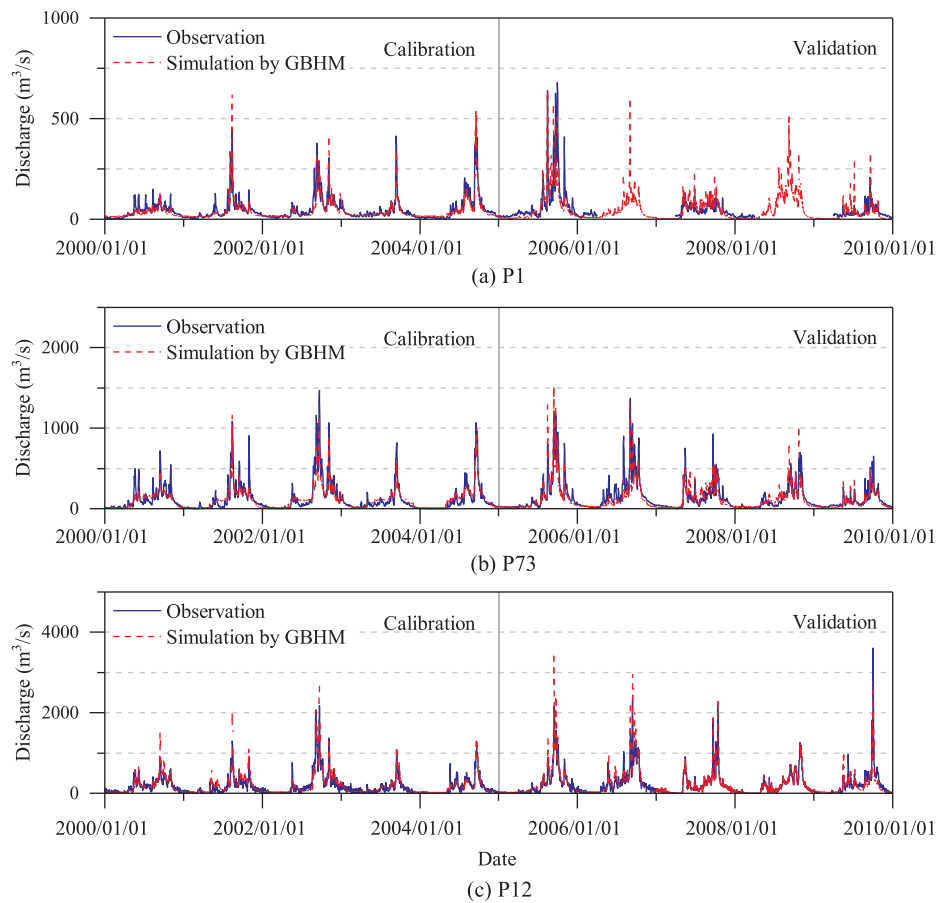


Fig. 3. Comparison of daily observed streamflow (blue) and simulated streamflow (red) by GBHM at P1 (a), P73 (b) and P12 (c) stations during the calibration and validation periods.

Table 3
Performances of the GBHM model during the calibration and validation period.

	Calibration period			Validation period		
	P1	P73	P12	P1	P73	P12
RMSE (m ³ /s)	25.04	81.89	107.50	38.97	83.08	124.73
NSE	0.79	0.80	0.79	0.68	0.78	0.84
PBIAS (%)	−9.6	−9.2	7.83	−4.3	−12.3	4.76
LNSE	0.58	0.78	0.62	0.74	0.80	0.81
PE (%)	−20.5	−20.8	−6.8	−6.8	−20.8	−11.2

15, and the texture feature of an image is represented by a 15-dimensional histogram vector. Detailed descriptions of the LBP method and its applications were given by Ojala et al. (1996), Khan et al. (2020), and Kaya et al. (2014).

Here, we use a spatial pyramid matching (SPM) strategy (Lazebnik et al., 2006) to capture the spatial layout of the texture and intensity features. For the three-level SPM used in this study, each image is split into 1 ($=2^0 \times 2^0$), 4 ($=2^1 \times 2^1$) and 16 ($=2^2 \times 2^2$) sub-images, and the spatial features (15-dimensional texture vector and 1-dimensional intensity vector) are independently extracted from each sub-image. All the sub-image features are concatenated to create a feature vector with a dimension of 336 ($=(1 + 4 + 16) \times (15 + 1)$). Finally, the spatial features of all 23 input variables (see Eq. (1)) are concatenated, and a feature space with a dimension of 8418 ($=23 \times 366$) is generated. We use the PCA method to further reduce the dimensionality of the feature space, and a 118-dimensional input feature space is finally obtained with a 95% explained variability. The details of the SPM strategy were reported by Lazebnik et al. (2006), and the detailed process of image

Table 4
The percentages of successful prediction of streamflow magnitude labels by RS-KNN.

Hydrological Stations	Successful prediction		
	P1	P73	P12
Training period	99.8%	99.6%	99.5%
Validation period	99.7%	99.6%	99.4%
Test period	91.5%	92.2%	93.1%

spatial feature extraction with the LBP-CV and SPM methods is described by Jiang et al. (2018).

3.4. Machine learning methods

We use two ML methods for regression and classification: ANN and RS-KNN classifiers. We split the synthetic data into training (80%) and validation (20%) sets to determine whether the model effectively learned the behavior of the GBHM. For all three gauges, the observation data from 1 January 2010 to 31 December 2016 were used to test the prediction capability of the models. The test period included extreme floods (2011).

3.4.1. Artificial neural network (ANN)

We use a three-layer ANN with initial conditions optimized by a genetic algorithm (GA) for hydrological simulations, in which GA is used to optimize the initial conditions. The detailed architecture of the ANN was presented by Yang et al. (2019), and more details of the GA were provided by Wang (1991).

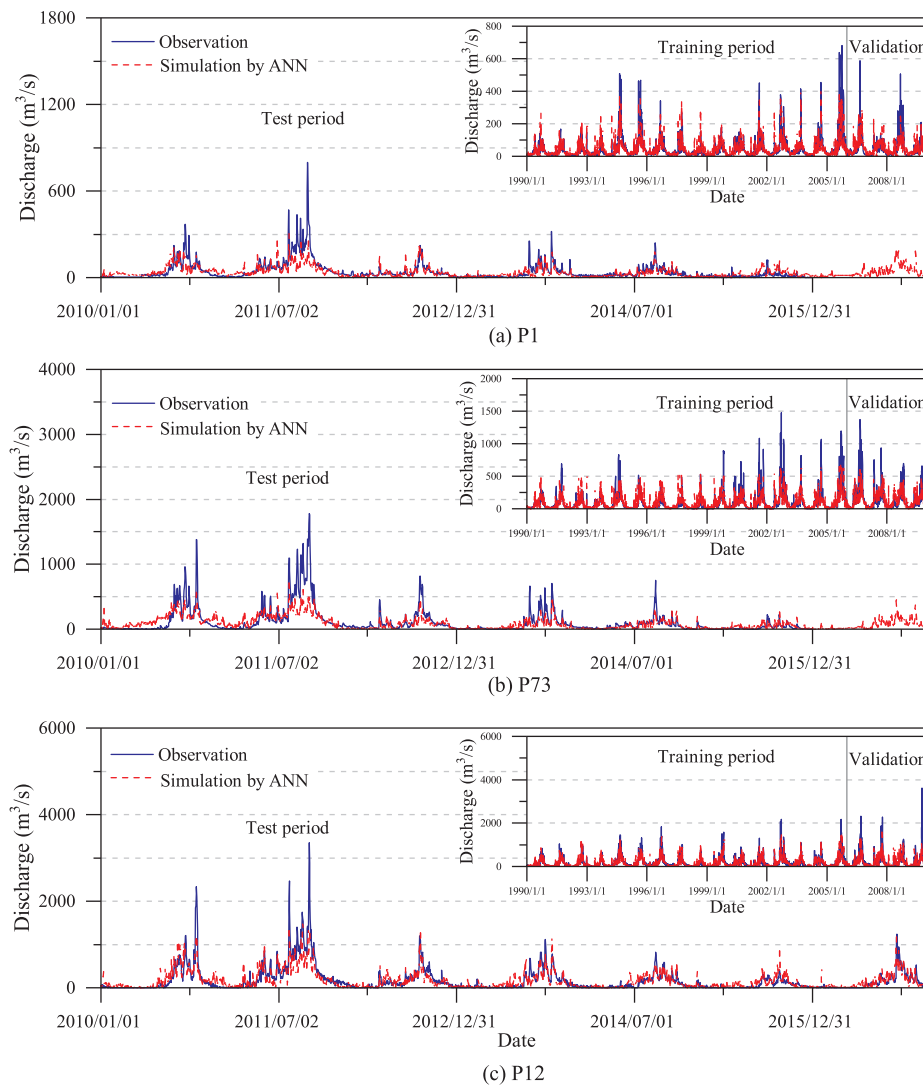


Fig. 4. Comparison of the observations (blue) and the simulated streamflow by ANN based on the historical data (red) at P1 (a), P73 (b) and P12 (c) stations during the training, validation and test periods.

In this study, we use the mean square error (MSE) as the loss function and the NSE as the objective function for the optimization of the initial conditions. We adopt the Levenberg-Marquardt (LM) method for neural network training. In addition, the *tanh* function is used as the activation function in this study, and the model is implemented with MATLAB software (Beale et al., 1992).

3.4.2. Ensemble *k*-nearest neighbor (RS-KNN) classifier

In this study, the streamflow data are separated into four sub-datasets, including those for small, medium, large and extraordinarily large flows, and the separation is determined based on the 80th, 95th and 98th percentiles of the flow frequency distribution curve obtained from historical observations (Sun et al., 2008). The 80th percentile is used to distinguish the small and medium flow, the 95th percentile is used to distinguish the medium and large flows, and the 98th percentile is used to distinguish the large and extraordinarily large flows. Under the separation criteria, the streamflow on each day is assigned a corresponding flow magnitude label.

We use the random subspace ensemble *k*-nearest neighbor classifier (RS-KNN) to predict the magnitude labels (i.e., small, medium, large or extraordinarily large) of the streamflow for each day in the test period. The *k*-nearest neighbor (KNN) (Aha et al., 1991) classifier is used to predict the membership of objects based on the closest training

instances in the feature space (Yang et al., 2012). The *k* closest neighbors of an instance are defined according to the Pearson correlation coefficient, and the label value of each instance is determined based on these neighbors with the majority vote principle (Yang et al., 2011). As a popular ensemble learning method, the random subspace algorithm includes bootstrapping and aggregation in the feature space (Chen et al., 2019). This approach can improve the efficiency and accuracy of classification compared to the values using a single classifier (Kuncheva et al., 2010). By combining RS and KNN, the RS-KNN classifier randomly divides the entire feature space into several subspaces, and the KNN classifier is trained for each subspace. The final ensemble decision is generated according to the majority vote principle. The details of the RS-KNN classifier were reported by Kuncheva et al. (2010), and the classifier was implemented with MATLAB software.

4. Results

In this section, we present the calibration and validation results of the GBHM and compare the performance of four data-driven models to demonstrate the effectiveness of the modelling framework proposed in this study. The prediction capability of the proposed model is examined with the data observed from 1 January 2010 to 31 December 2016 in the study area.

Table 5
Comparison of four models for understanding the effectiveness of different modeling approaches.

		Training period		Validation period			Test period		
		P1		P73	P12	P1	P73	P12	
(1) ANN	RMSE (m ³ /s)	33.95		91.40	120.48	39.67	129.90	190.95	43.99
	NSE	0.67		0.58	0.74	0.45	0.29	0.60	0.51
	PBIAS (%)	1.0		1.3	0.3	12.2	23.7	-7.1	-10.2
	LNSE	0.65		0.65	0.68	0.47	0.21	0.68	0.51
	PE (%)	-23.4		-28.3	-29.7	-38.4	-43.7	-33.9	-50.6
(2) GBHM-ANN	RMSE (m ³ /s)	42.75		81.42	180.28	46.53	87.86	194.01	39.42
	NSE	0.73		0.75	0.76	0.44	0.52	0.52	0.67
	PBIAS (%)	6.6		-1.0	-2.1	13.8	6.9	12.4	12.0
	LNSE	0.68		0.77	0.76	0.53	0.59	0.55	0.56
	PE (%)	-16.2		-19.6	-24.2	-24.9	-17.2	-24.1	-35.4
(3) GBHM-ANN-CA	RMSE (m ³ /s)	20.14		41.34	93.75	18.12	37.68	80.97	24.51
	NSE	0.94		0.94	0.94	0.92	0.91	0.92	0.87
	PBIAS (%)	-6.4		-5.6	-1.7	-8.8	-5.4	-1.6	8.4
	LNSE	0.89		0.88	0.91	0.78	0.79	0.83	0.79
	PE (%)	-5.7		-9.7	-7.7	-3.1	2.2	-14.1	-7.1
(4) GBHM-ANN-CA-CV	RMSE(m ³ /s)	12.48		25.63	52.55	11.60	22.24	51.58	23.00
	NSE	0.98		0.98	0.98	0.97	0.97	0.97	0.89
	PBIAS (%)	0.6		0.8	0.2	-0.6	0.3	-1.2	11.2
	LNSE	0.95		0.94	0.93	0.94	0.91	0.89	0.75
	PE (%)	-0.5		-1.3	-1.9	-4.4	-1.4	-10.4	4.6

(1) ANN is the simple artificial neural network.

(2) GBHM-ANN is the ANN model driven by GBHM generated synthetic data.

(3) GBHM-ANN-CA is the ANN coupling with categorization approach driven by GBHM generated synthetic data.

(4) GBHM-ANN-CA-CV is the ANN coupling with categorization approach (CA) and computer vision (CV) driven by GBHM generated synthetic data.

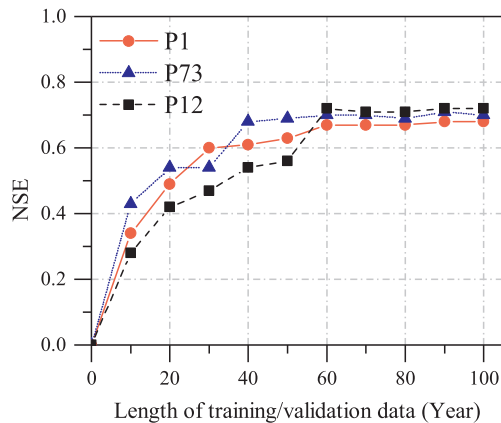


Fig. 5. The impact of the synthetic data lengths on the performance of GBHM-ANN at the three gauges. The performance is evaluated based on the NSE value during the test periods, and the synthetic data are used for model training and validation.

4.1. Calibration and validation of GBHM

We first calibrate and validate the GBHM using data from three stations. Fig. 3 and Table 3 show the streamflow simulations for three gauges obtained with the GBHM in the calibration and validation periods. At gauge P1, the model achieves a high NSE value of 0.79 and a slight underestimation with a PBIAS of -9.6% for the daily discharge in the calibration period; additionally, the NSE is 0.68 and the PBIAS value is -4.3% in the validation period. At gauge P73, the model achieves a high NSE value of 0.80 and a slight underestimation with a PBIAS of -9.2% in the calibration period; in the validation period, the NSE is 0.78, and PBIAS is -12.3%. At gauge P12, the NSE values are 0.79 and 0.84 and the PBIAS values are 7.83% and 4.76% in the calibration and validation periods, respectively. The results show that the calibrated GBHM model can effectively describe the hydrological processes in the Ping River Basin and be used to generate streamflow time

series.

4.2. Results of the synthetic studies

To evaluate how three proposed modeling approaches are effective, we use the simulation results of the simple ANN by using limited observations as the baseline, and design three experiments, including streamflow simulations by GBHM-ANN, GBHM-ANN-CA and GBHM-ANN-CA-CV with synthetic data. Before showing the results of the four models, we summarize the performance of the RS-KNN classifier in Table 4. During the training, validation and test periods, the RS-KNN performs well for the three gauges, with correct percentages of more than 90%. These results indicate that the classifier can accurately predict the magnitude labels of streamflow according to precipitation information.

As a baseline model, the simple ANN is trained and validated with 20-year observations. We consider the number of hidden neurons as a tuning parameter and select it from a preset of ranges between 5 and 30. For each value, we run for 20 times and the maximum NSE among the 20 times is taken as the representative value. We use the same tuning method for all the ML models in this study. After adjusting hyperparameters, we use architectures with 12 hidden neurons for gauge P1, 10 hidden neurons for gauge P12, and 15 hidden neurons for gauge P73. The results are shown in Fig. 4 and Table 5. The simple ANN adequately simulates the streamflow at gauge P12, but false simulations are generated at gauges P1 and P73. The difference in performance for the three gauges indicates that the ANN model lacks robustness, which may be caused by ignoring the spatial variability of predictors. In addition, the ANN performs poorly under both low flow and peak flow conditions and significantly underestimates the peak flows at all three gauges, likely due to the inadequacy of training data. For the extreme flood in 2011 (Mateo et al., 2015), the ANN model is unable to simulate the event at all three gauges and significantly underestimates the peak flow, with PE values greater than 49%. Although the ANN is an easy-to-use method that requires only several seconds to run 10-year simulations, it must be improved to be a surrogate model of the GBHM.

To overcome the problem of inadequate training data, physically

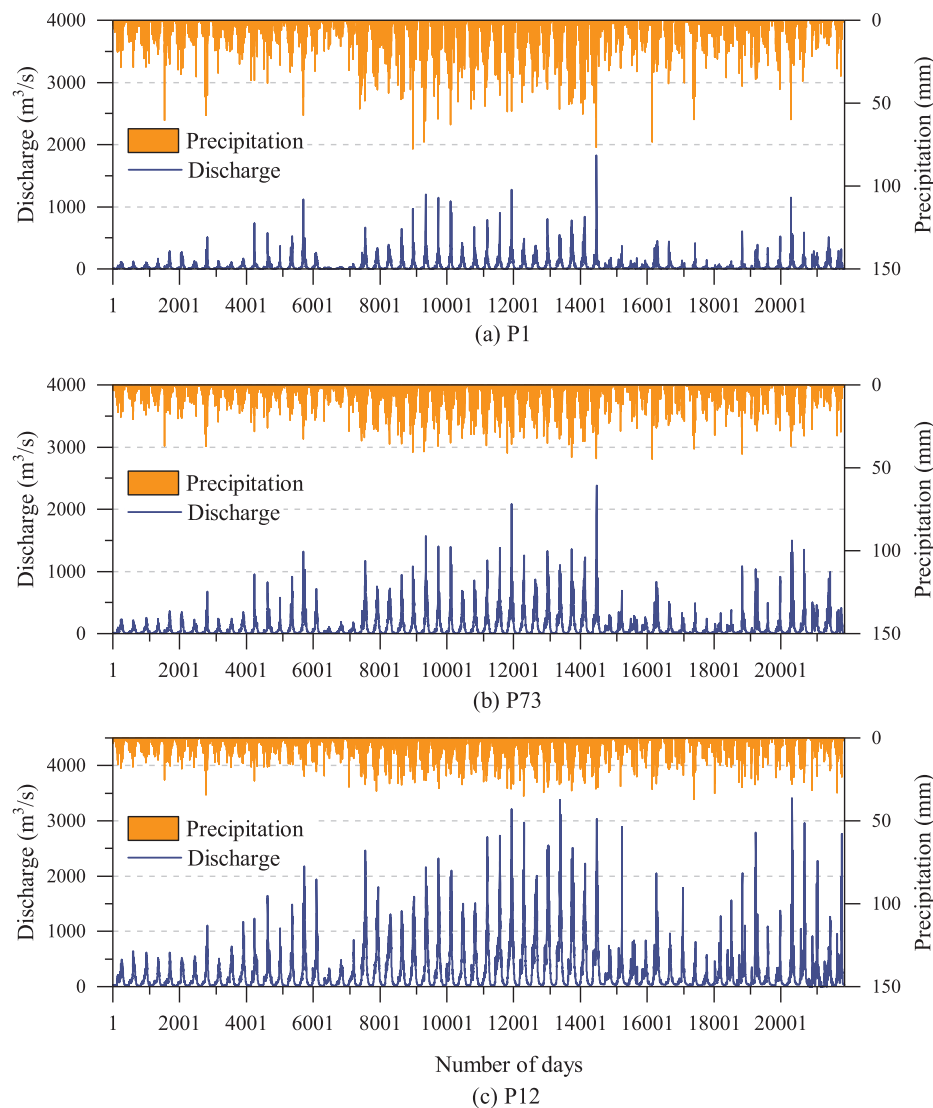


Fig. 6. A 60-year areal mean precipitation data generated by stochastic rainfall generator and the corresponding streamflow data generated by the calibrated GBHM at P1 (a), P73 (b) and P12 (c) stations. The orange bar is the daily areal mean precipitation, and the blue solid line is the daily streamflow data generated by GBHM.

based synthetic data are generated by the GBHM and SRG for model training and validation. To determine the suitable lengths of synthetic time series data for model training and validation, we calculate the performance (i.e. *NSE* during the test period) of GBHM-ANN using different lengths of synthetic data. The synthetic data are split into training (80% of data for initially fitting) and validation (20% of data for hyper-parameter tuning) sets according to a commonly adopted “80/20” split rule (Yang et al., 2016), and all models are tested with the climate forcing observations and corresponding GBHM-generated streamflow data from 1 January 2010 to 31 December 2016. As shown in Fig. 5, it can be seen that the simulations become stable with an acceptable accuracy ($NSE \approx 0.7$) when the data lengths were more than 60 years. The 60-year synthetic data are shown in Fig. 6.

Here, we emphatically introduce the simulation results of GBHM-ANN with 60-year synthetic data. It should be clarified that the performance of the GBHM-ANN is evaluated based on a comparison of the GBHM-ANN simulations and synthetic runoff data. The architecture of the GBHM-ANN includes 25 hidden neurons for gauge P1, 24 hidden neurons for gauge P12, and 28 hidden neurons for gauge P1 after adjusting hyperparameters with the same parameter adjustment method as that used by simple ANN. The results of GBHM-ANN are shown in Fig. 7 and Table 5. The model performs satisfactorily at all three gauges during the test period, while an unbalanced performance of the model

at the three gauges still exists with a worse simulation at gauge P1. For the extreme flood in 2011, the GBHM-ANN provides a better flood simulation result with higher *NSE* and lower *PBIAS* values, but significantly underestimates the peak flow at the three gauges with large *PE* values (more than 30%). Although the significant peak flow underestimations have been greatly mitigated, the peak flow simulation is still unsatisfactory with high *PE* values; this issue must be addressed in the next modeling step.

To improve the ability of GBHM-ANN to simulate high flows, a categorization approach is used, and GBHM-ANN-CA is developed. The architecture of GBHM-ANN-CA includes 12 hidden neurons for small-magnitude modules, 10 hidden neurons for medium-magnitude modules, 5 hidden neurons for large-magnitude modules and 5 hidden neurons for extraordinarily large-magnitude modules at all three gauges after the adjustment of hyperparameters with the same parameter adjustment method as simple ANN. Fig. 8 and Table 5 show the simulation results and performance of GBHM-ANN-CA. With the support of CA, the performance of the model can be judged as satisfactory at all three gauges, and the simulation of peak flow and low flow has been improved greatly. In terms of the extreme flood in 2011, GBHM-ANN-CA provides a satisfactory result for peak flows, with a low *PE* (less than 15%), but significantly overestimates the peak flow at gauge P73, with a high *PE* value of 27.1%. This overestimation at P73 may be

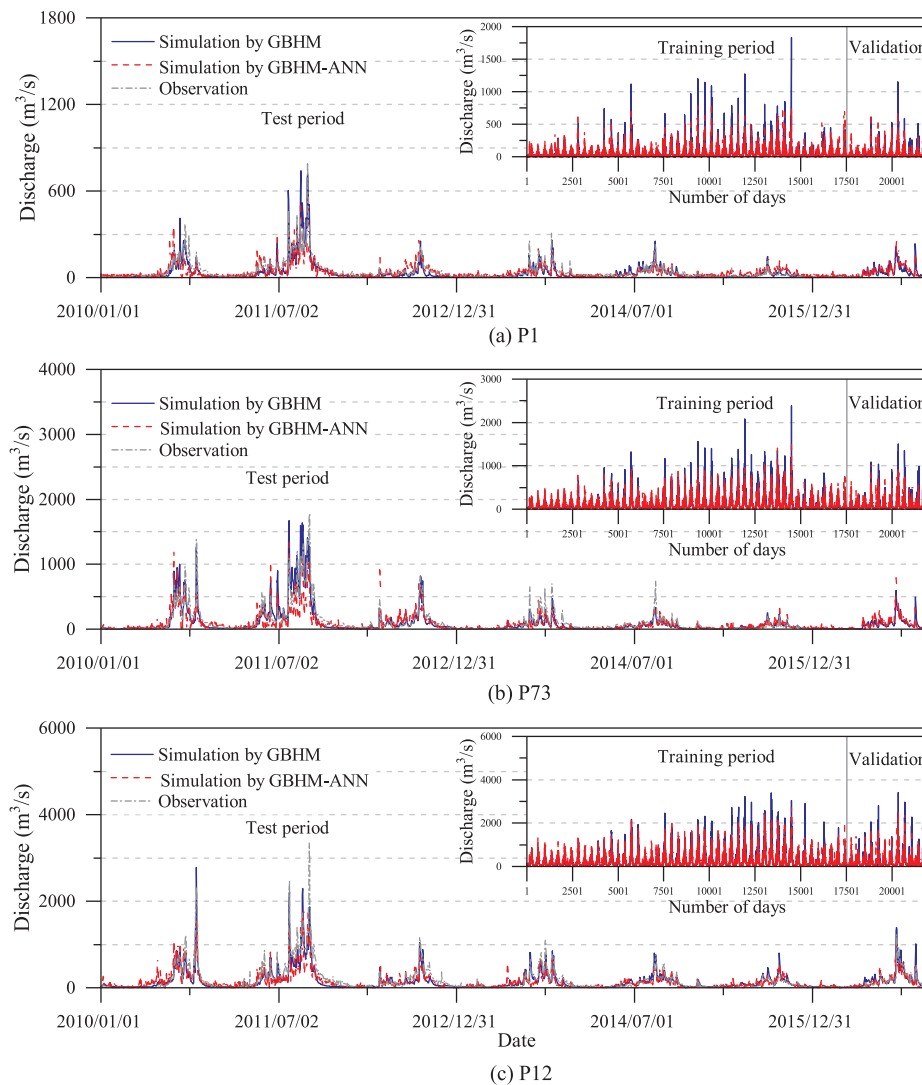


Fig. 7. The effectiveness of the guidance of the physically-based hydrological model (GBHM) in data-insufficient basins: comparison of daily simulated streamflow by GBHM (blue) and the simulated streamflow by the ANN driven by GBHM generated synthetic data (GBHM-ANN) (red) at P1 (a), P73 (b) and P12 (c) stations during the training, validation and test periods.

caused by the sparsity of rainfall gauges. In addition, a relatively large difference between peak flow simulations at three gauges still exists and needs to be improved in the next modeling step.

To improve the robustness of GBHM-ANN-CA, LBP-CV is used to extract the spatial features, and GBHM-ANN-CA-CV is proposed. The architectures of the four modules have 25 hidden neurons in the hidden layer after hyperparameter adjustment with the same parameter adjustment method as simple ANN. As shown in Fig. 9 and Table 5, the model can learn the hydrological behavior of the GBHM with a good performance, and the model performance levels for the three gauges are satisfactory, with *NSE* values greater than 0.85 and *PBIAS* values less than 15%. In terms of model performance for low flow and peak flow simulations, the *LNSE* values are all greater than 0.65 and the *PE* values are all less than 15% at three gauges. For the extreme flood in 2011, GBHM-ANN-CA-CV exhibit satisfactory performances, with high *NSE* (greater than 0.8) and low *PBIAS* (less than 15%) values.

4.3. Results of a real case study

After examining the model performance in learning the hydrological behavior of the GBHM, the GBHM-ANN-CA-CV model can be regarded as the best surrogate model for the GBHM, and the prediction capability

of the model needs to be investigated with real data. Therefore, we compare the GBHM-ANN-CA-CV simulations with observations at the three gauges. As shown in Fig. 10 and Table 6, since GBHM-ANN-CA-CV is driven by GBHM-generated synthetic data, its prediction capability will not exceed that of the GBHM; thus, we also compare the observations and simulations of the GBHM during the test period.

The results show that both the GBHM and GBHM-ANN-CA-CV yield satisfactory performances at the three gauges, although the prediction accuracy of GBHM-ANN-CA-CV is lower than that of GBHM, with *NSE* values that are approximately 10% higher. In low flow prediction, the prediction accuracy of GBHM-ANN-CA-CV is also lower than that of the GBHM, with *LNSE* values that are approximately 15% higher. In terms of peak flow prediction, there are slight underestimations in the GBHM results and slight overestimations in the GBHM-ANN-CA-CV results at P1 and P73, and there are significant underestimations in both the GBHM and GBHM-ANN-CA-CV results at P12.

In terms of the computing cost, the total computational time is greatly reduced by ML. It takes on the order of tens of seconds to run the 10-year simulation using the GBHM-ANN-CA-CV model, and it takes on the order of tens of hours for the 10-year simulation using the GBHM model. The considerable improvement in the computational time can be useful for practical water resource management, especially for real-

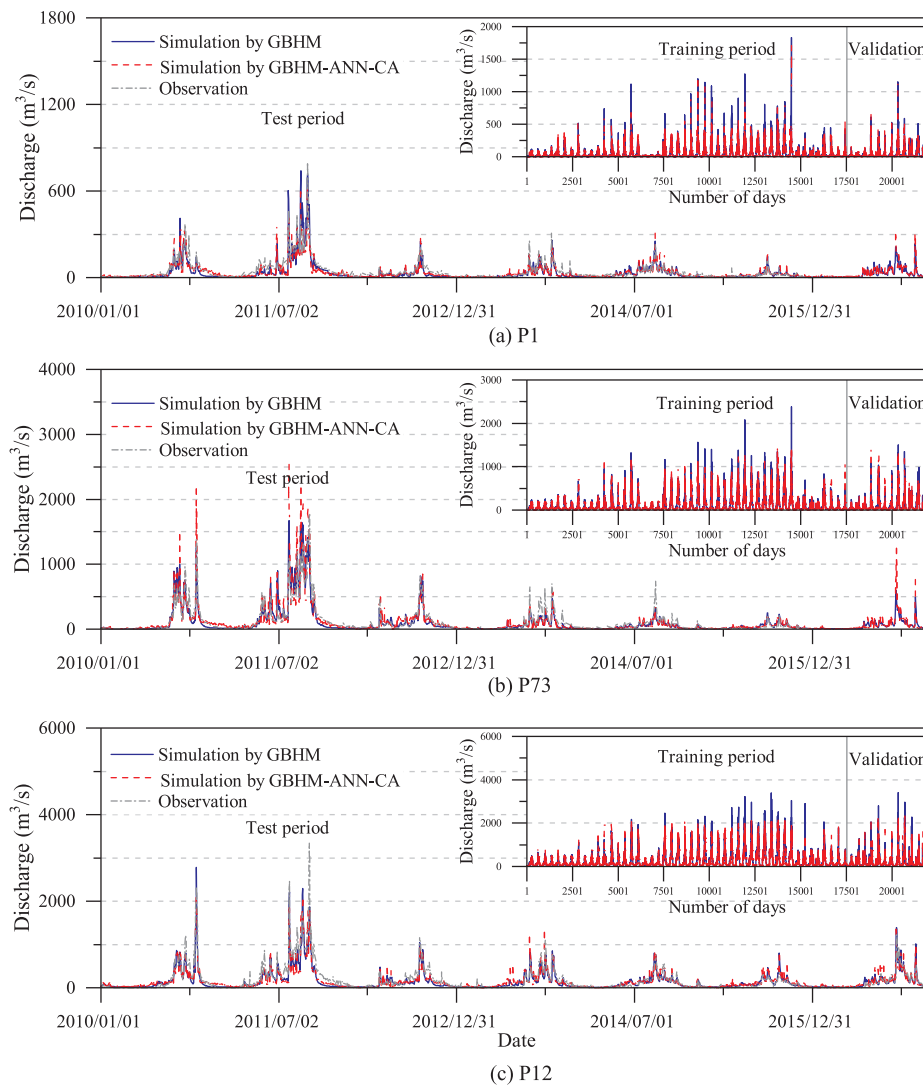


Fig. 8. The effectiveness of the categorization modeling approach: comparison of daily simulated streamflow by GBHM (blue) and the simulated streamflow by the ANN coupling with a categorization approach (CA) driven by GBHM generated synthetic data (GBHM-ANN-CA) (red) at P1 (a), P73 (b) and P12 (c) stations during the training, validation and test periods.

time streamflow forecasting and flash flood warnings.

5. Discussion

5.1. Comparison of different modeling approaches

1) Effectiveness of the hydrological mechanisms-guided modeling approach

To demonstrate the effectiveness of hydrological mechanism-guided modeling approaches, we compare the performances of the GBHM-ANN with those of a simple ANN. GBHM-ANN, which adopts sufficient synthetic data for training and validation, performs better at all three gauges. The *NSE* values of the simulation results at the three gauges are improved by an average of 23% during the test period. Specifically, with sufficient training data, the significant underestimation of peak flow has been alleviated. This comparison suggests that physically based hydrological models can greatly improve the performance of data-driven models when historical observations are limited.

2) Effectiveness of the categorization approach

We compare the performance of GBHM-ANN-CA and GBHM-ANN to demonstrate the effectiveness of the categorization approaches. GBHM-ANN-CA is superior since it has much higher *NSE* and lower *PBIAS* values and a better performance under both low flow and peak flow conditions. By using separate sub-datasets for the simulation, the CA avoids the insufficient training of high and low flows caused by their low frequencies in flow series, and thus improves the *NSE* values by an average of 27% during the test period at the three gauges. The results show that the CA based GBHM-ANN model can significantly improve the effectiveness of the peak flow simulation.

3) Effectiveness of the spatial information extracted by the LBP-CV method

Compared with the robust performance of GBHM-ANN-CA-CV for peak flow simulation, the *PE* values of the GBHM-ANN-CA simulation show a significant overestimation for gauge P73 and a slight underestimation for gauges P1 and P12. Considering the spatial variability of the predictors, the LBP-CV method solves the performance differences of GBHM-ANN-CA at the three gauges, and improves the *NSE* values of all the three gauges by an average of 2% during the test period. The results highlight that the spatial feature information extracted by the

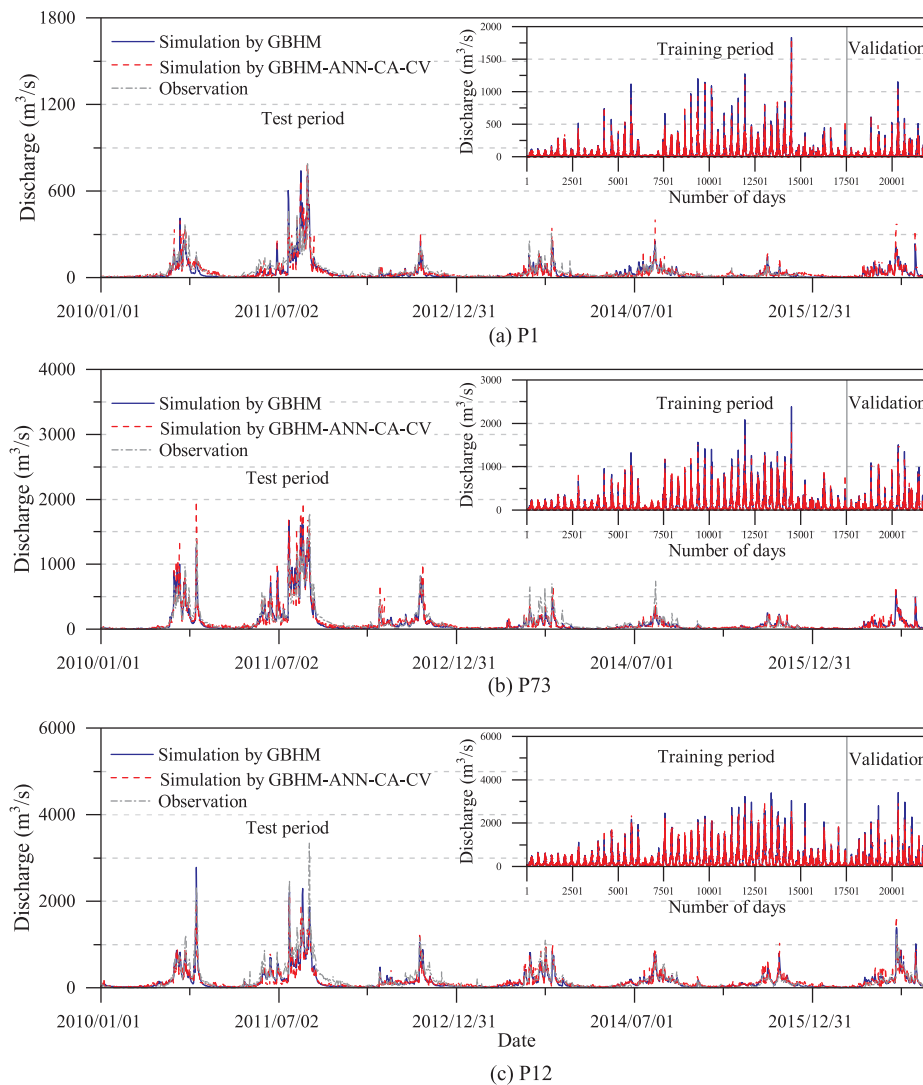


Fig. 9. The Effectiveness of spatial information extracted by LBP based computer vision: comparison of daily simulated streamflow by GBHM (blue) and the simulated streamflow by the ANN coupling with a categorization approach (CA) and computer version (CV) driven by GBHM generated synthetic data (GBHM-ANN-CA-CV) (red) at P1 (a), P73 (b) and P12 (c) stations during the training, validation and test periods.

LBP-CV approach can improve the simulation results, especially for peak flow simulations.

5.2. Simulation uncertainties of GBHM-ANN-CA-CV

We argue that the simulation uncertainties of the GBHM-ANN-CA-CV model mainly come from the GBHM model and the ML methods adopted in the model. The error of the GBHM mainly comes from the uncertainties associated with the model parameters, model structure and meteorological inputs (Li et al., 2012). Since the soil depth and soil types were obtained from the HWS dataset with a relatively low spatial resolution, the heterogeneity of the soil information may not be fully captured at the regional scale, and thus, the estimated soil parameters may be influenced by large uncertainties. Limited by the understanding of hydrological processes, the model structure cannot fully describe real hydrological processes with high accuracy, and this limitation can result in uncertainties. Additionally, limited by the number of meteorological stations in the study area, the spatial interpolation of the meteorological predictors is a relatively uncertain process and may increase the simulation error. The error in ML mainly comes from the uncertainties of label prediction with the RS-KNN and the streamflow prediction with the ANN.

In the future, since remote sensing can effectively capture the spatial heterogeneity of meteorological variables, remote sensing data can be used in the model, thus reducing the model uncertainties. In addition, compared with the physically based hydrological model, the model proposed in this study is unable to simultaneously provide discharge simulations for different catchments. Future studies need to explore how to develop a fully distributed data-driven hydrological model that not only considers the spatial heterogeneity of predictors but also provides distributed hydrological simulations.

6. Conclusion

This study proposed three approaches to improve the prediction capability of machine learning-based hydrological models and developed a physical process and ML combined hydrological model for rainfall-runoff simulation in data-insufficient watersheds. The model performance in learning the hydrological behaviors of the physically distributed hydrological model was evaluated with a synthetic dataset, and the prediction capability was investigated using real data. According to the results, we developed the following conclusions.

- (1) Driven by the synthetic data generated by a distributed

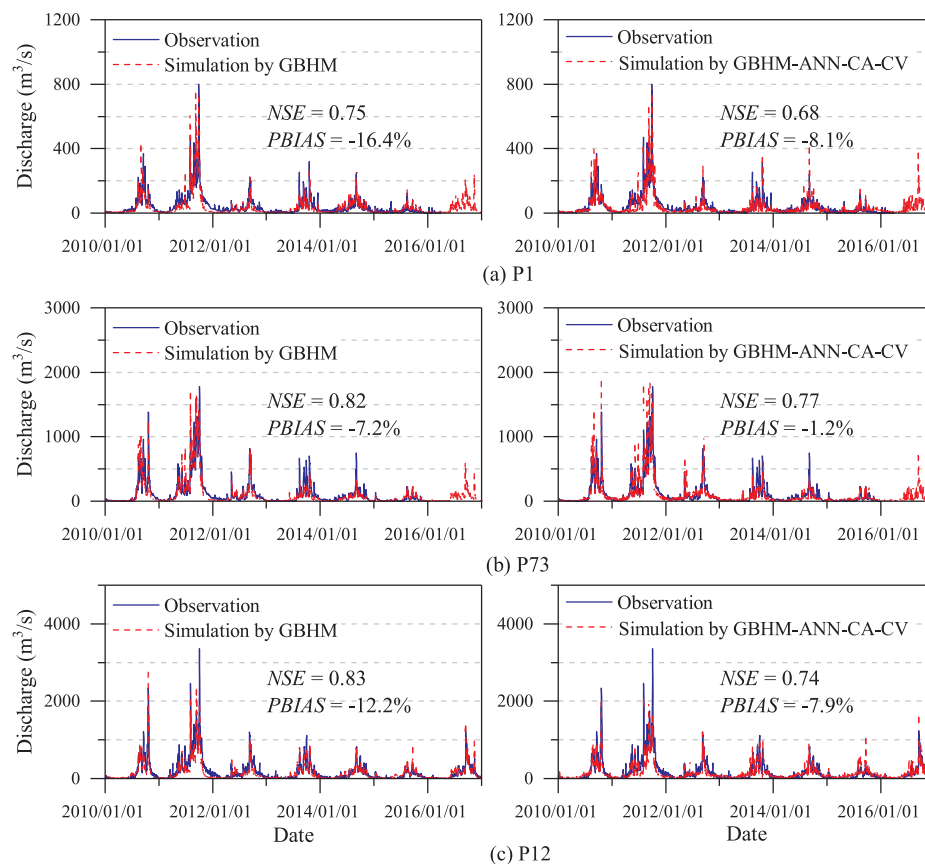


Fig. 10. The prediction capability of the proposed model: comparison of streamflow between the observations (blue) and the simulation of GBHM (red), and the comparison of streamflow between the observations (blue) and the simulation of GBHM-ANN-CA-CV (red) at P1 (a), P73 (b) and P12 (c) stations.

Table 6

Prediction accuracies of the physical process and machine learning combined hydrological model (GBHM-ANN-CA-CV) and GBHM in the test period from 1 January 2010 to 31 December 2016.

Model	GBHM			GBHM-ANN-CA-CV		
	P1	P73	P12	P1	P73	P12
RMSE (m³/s)	28.63	79.70	117.95	33.28	91.26	144.30
NSE	0.75	0.82	0.83	0.68	0.77	0.74
PBIAS (%)	-16.4	-7.2	-12.2	-8.1	-1.2	-7.9
LNSE	0.71	0.84	0.61	0.63	0.74	0.51
PE (%)	-2.2	-0.4	-19.2	2.9	14.7	-30.7

hydrological model, the performance of the ML-based data-driven model was greatly improved for daily runoff simulations in watersheds with limited historical runoff observations. By using the long-term time series generated by the GBHM and SRG, the insufficient training problem that often occurs in daily streamflow simulations due to data scarcity can be overcome, and the NSE values were improved by an average of 23% during the test period.

- (2) The streamflow categorization approach greatly improved the peak flow simulation capability of the data-driven hydrological model, and the NSE values were improved by 27% on average, during the test period. This approach is potentially useful for real-time water resource management, especially for flood and drought prediction.
- (3) The CV methods used in this study can effectively capture the spatial heterogeneity of predictors, which can significantly improve the robustness of data-driven hydrological models for rainfall-runoff simulations in large river basins.
- (4) The model developed in this study can effectively learn the hydrological behaviors of physically based distributed hydrological

models (e.g., the GBHM) and can be applied for long-term daily streamflow simulation and prediction in large data-insufficient catchments.

In addition, with the rapid development of computer technology and artificial intelligence, the modelling framework proposed in this study can be improved in the future with enhancements in physically distributed hydrological models, CV methods, categorization methods and ML.

CRediT authorship contribution statement

Shuyu Yang: Conceptualization, Methodology, Data curation, Writing - original draft, Writing - review & editing. **Dawen Yang:** Supervision, Conceptualization, Methodology, Writing - review & editing, Funding acquisition. **Jinsong Chen:** Supervision, Methodology, Writing - review & editing. **Jerasorn Santisirisomboon:** Resources. **Weiwei Lu:** Investigation. **Baoxu Zhao:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was financially supported by the National Natural Science Foundation of China (project Nos. 41661144031). The authors would like to thank the Royal Irrigation Department of Thailand and Department of Land Development of Thailand for providing the

historical observations, including the daily runoff data at three gauges (P.1, P73 and P12), the daily precipitation, the daily air temperature (mean, maxima and minima), the daily mean relative humidity, the daily sunshine duration and the land use map. In addition, the authors also thank the researches and their teams for the datasets used in this study. HWS dataset is available at <http://www.fao.org/soils-portal/soil-survey/soil-maps-and-databases/harmonized-world-soil-database-v12/en/>. The SRTM 90m DEM dataset is available at <http://srtm.csi.cgiar.org/srtmdata/>. The GIMMS NDVI3g is available at <https://ecocast.arc.nasa.gov/data/pub/gimms/3g.v1/>. The GLDAS dataset is available at <https://disc.gsfc.nasa.gov/datasets?keywords=GLDAS&page=1>.

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